

Overview of Image super-resolution

Subject: Image super-resolution

1.

Probing near-field electromagnetic disturbance Super-resolution microscopy is generally discussed within the realm of conventional optical imagery. However, modern technology allows the probing of electromagnetic disturbance within molecular distances of the source, which has superior resolution properties. See also evanescent waves and the development of the new super lens.

2.

Technical implementations There are many both single-frame and multiple-frame variants of SR. Multiple-frame SR uses the sub-pixel shifts between multiple low resolution images of the same scene. It creates an improved resolution image fusing information from all low resolution images, and the created higher resolution images are better descriptions of the scene. Single-frame SR methods attempt to magnify the image without producing blur. These methods use other parts of the low resolution images, or other unrelated images, to guess what the high-resolution image should look like. Algorithms can also be divided by their domain: frequency or space domain. Originally, super-resolution methods worked well only on grayscale images, but researchers have found methods to adapt them to color camera images. Recently, the use of super-resolution for 3D data has also been shown.

3.

Sub-pixel image localization The location of a single source can be determined by computing the "center of gravity" (centroid) of the light distribution extending over several adjacent pixels (see figure on the left). Provided that there is enough light, this can be achieved with arbitrary precision, very much better than pixel width of the detecting apparatus and the resolution limit for the decision of whether the source is single or double. This technique, which requires the presupposition that all the light comes from a single source, is at the basis of what has become known as super-resolution microscopy, e.g. stochastic optical reconstruction microscopy (STORM), where fluorescent probes attached to molecules give nanoscale distance information. It is also the mechanism underlying visual hyperacuity.

4.

Multiplexing spatial-frequency bands An image is formed using the normal passband of the optical device. Then, some known light structure (for example, a set of light fringes) is superimposed on the target. The image now contains components resulting from the combination of the target and the superimposed light structure, e.g. moiré fringes, and carries information about target detail which simple unstructured illumination does not. The "superresolved" components, however, need disentangling to be revealed. For an example, see structured illumination (figure to left).

5.

Super-resolution imaging (SR) is a class of techniques that improve the resolution of an imaging system. In optical SR the diffraction limit of systems is transcended, while in geometrical SR the

resolution of digital imaging sensors is enhanced. In some radar and sonar imaging applications (e.g. magnetic resonance imaging (MRI), high-resolution computed tomography), subspace decomposition-based methods (e.g. MUSIC) and compressed sensing-based algorithms (e.g., SAMV) are employed to achieve SR over standard periodogram algorithm. Super-resolution imaging techniques are used in general image processing and in super-resolution microscopy.

6.

Research Research into using neural network computing to perform super-resolution image construction. For example, deep convolutional networks were used to generate a 1500x scanning electron microscope image from a 20x microscopic image of pollen grains. However, while this technique can increase the information content of an image, there is no guarantee that the upscaled features actually exist in the original image. For this reason deep convolutional upscalers are not appropriate for applications involving ambiguous inputs where the presence or absence of a single feature is critical. Hallucinated details in images taken for medical diagnosis, as an example, could be problematic.

7.

Other related work Curtis, Craig H.; Milster, Tom D. (October 1992). "Analysis of Superresolution in Magneto-Optic Data Storage Devices". *Applied Optics*. 31 (29): 6272–6279. Bibcode:1992ApOpt..31.6272M. doi:10.1364/AO.31.006272. PMID 20733840. Zalevsky, Z.; Mendlovic, D. (2003). *Optical Superresolution*. Springer. ISBN 978-0-387-00591-1. Caron, J.N. (September 2004). "Rapid supersampling of multiframe sequences by use of blind deconvolution". *Optics Letters*. 29 (17): 1986–1988. Bibcode:2004OptL...29.1986C. doi:10.1364/OL.29.001986. PMID 15455755. Clement, G.T.; Huttunen, J.; Hynynen, K. (2005). "Superresolution ultrasound imaging using back-projected reconstruction". *Journal of the Acoustical Society of America*. 118 (6): 3953–3960. Bibcode:2005ASAJ..118.3953C. doi:10.1121/1.2109167. PMID 16419839. Geisler, W.S.; Perry, J.S. (2011). "Statistics for optimal point prediction in natural images". *Journal of Vision*. 11 (12): 14. doi:10.1167/11.12.14. PMC 5144165. PMID 22011382. Cheung, V.; Frey, B. J.; Jojic, N. (20–25 June 2005). Video epitomes (PDF). *Conference on Computer Vision and Pattern Recognition (CVPR)*. Vol. 1. pp. 42–49. doi:10.1109/CVPR.2005.366. Bertero, M.; Boccacci, P. (October 2003). "Super-resolution in computational imaging". *Micron*. 34 (6–7): 265–273. doi:10.1016/s0968-4328(03)00051-9. PMID 12932769. Borman, S.; Stevenson, R. (1998). "Spatial Resolution Enhancement of Low-Resolution Image Sequences – A Comprehensive Review with Directions for Future Research" (Technical report). University of Notre Dame. Borman, S.; Stevenson, R. (1998). Super-resolution from image sequences — a review (PDF). *Midwest Symposium on Circuits and Systems*. Park, S. C.; Park, M. K.; Kang, M. G. (May 2003). "Super-resolution image reconstruction: a technical overview". *IEEE Signal Processing Magazine*. 20 (3): 21–36. Bibcode:2003ISPM...20...21P. doi:10.1109/MSP.2003.1203207. S2CID 12320918. Farsiu, S.; Robinson, D.; Elad, M.; Milanfar, P. (August 2004). "Advances and Challenges in Super-Resolution". *International Journal of Imaging Systems and Technology*. 14 (2): 47–57. doi:10.1002/ima.20007. S2CID 12351561. Elad, M.; Hel-Or, Y. (August 2001). "Fast Super-Resolution Reconstruction Algorithm for Pure Translational Motion and Common Space-Invariant Blur". *IEEE Transactions on Image Processing*. 10 (8): 1187–1193. Bibcode:2001ITIP...10.1187E. CiteSeerX 10.1.1.11.2502. doi:10.1109/83.935034. PMID 18255535. Irani, M.; Peleg, S. (June 1990). Super Resolution From Image Sequences (PDF). *International Conference on Pattern Recognition*. Vol. 2. pp. 115–120.

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